Ertel Potential Vorticity on Sigma Levels

PV_Ertel_Computation Project

v0.4.1 — June 2026

1 Ertel PV in Height Coordinates

Ertel's potential vorticity (1) in height coordinates (x, y, z):

$$\mathrm{EPV} = \frac{1}{\rho} (\vec{\omega}_a \cdot \nabla\theta) \tag{1}$$

where $\vec{\omega}_a = \nabla \times \vec{u} + 2\vec{\Omega}$ (absolute vorticity), $\theta = T(p_0/p)^{R_d/c_p}$ (dry potential temperature, $p_0 = 1000\,\mathrm{hPa}$).

Horizontal derivatives use spectrally accurate spherical harmonic transforms via `pvtend.sh_ops`, eliminating polar singularities.

2 Sigma-Coordinate Formulation

Terrain-following sigma: $\sigma = p/p_s$ where $p_s(x, y)$ is surface pressure [Pa]. Mimics MPAS-Atmosphere's native vertical coordinate.

Important distinction:

- $p_0 = 100{,}000\,\mathrm{Pa}$ = constant — thermodynamic reference for θ
- $p_s(x, y)$ = varying 2-D field — surface pressure for $\sigma = p/p_s$

These are separate quantities; p_0 never appears in the sigma PV formula.

Coordinate transformation $\partial/\partial p \to (1/p_s)\partial/\partial\sigma$:

$$\boxed{\mathrm{PV}_\sigma = -\frac{g}{p_s}\left[\frac{\partial v}{\partial\sigma}\frac{\partial\theta}{\partial x}\bigg|_\sigma - \frac{\partial u}{\partial\sigma}\frac{\partial\theta}{\partial y}\bigg|_\sigma + (f + \zeta_\sigma)\frac{\partial\theta}{\partial\sigma}\right]} \tag{2}$$

Analytic equivalence: Ertel PV is coordinate-invariant: $\mathrm{PV}_\sigma(x, y, \sigma) = \mathrm{PV}_p(x, y, p = \sigma p_s)$.

Simplified form: $\mathrm{PV}_{\sigma,\mathrm{simple}} = -(g/p_s)\,(f + \zeta_\sigma)\,\partial\theta/\partial\sigma$.

3 Numerical Discretization

3.1 Vertical Derivatives

MPAS-style centred/sided differences. Coordinate-array ordering is auto-detected (ascending or descending). Boundary derivatives follow MPAS `calc_vertDeriv_one`.

3.2 Sigma Interpolation

Fields are interpolated from pressure to sigma levels via $\log p$-linear interpolation. Values where σp_s falls outside $[p_{\min}, p_{\max}]$ are filled from the nearest valid level.

3.3 11-Level Essential Sigma Grid

The sigma grid uses 11 essential levels spanning the boundary layer through the lower stratosphere — all synoptically relevant and all within ERA5 data range globally ($\sigma_{\min} \cdot \min(p_s) > p_{\min}$):

$$\sigma_k \in \{1.000,\ 0.925,\ 0.850,\ 0.700,\ 0.600,\ 0.500,\ 0.400,\ 0.300,\ 0.250,\ 0.200,\ 0.100\} \qquad (3)$$

At sea level ($p_s \approx 101325\,\mathrm{Pa}$): $p \approx \sigma_k \cdot 101325\,\mathrm{Pa}$. Terrain-following: $p = \sigma_k \cdot p_s(x, y)$.

This grid deliberately excludes $\sigma < 0.1$ (above $\sim$100 hPa) — mid-stratospheric levels where $\partial\theta/\partial\sigma$ becomes extremely large, producing unrealistic PV extremes ($\pm 10^5$ PVU). The 11-level grid produces PV in the range $[-51, 71]$ PVU, consistent with ERA5 native PV $[-62, 111]$ PVU.

4 Validation

4.1 ERA5 (2025-01-08 00Z, 11 σ levels)

Level	RMSE (PVU)	Correlation
$\sigma = 0.850$ (nom. 861 hPa)	0.30	0.92
$\sigma = 0.500$ (nom. 507 hPa)	0.37	0.94
$\sigma = 0.250$ (nom. 253 hPa)	1.12	0.97

All NaN fractions: 0 % (computed PV), 5.3 % (ERA5 native PV has missing values at some levels). PV range matches ERA5 native PV without the stratospheric blow-up of the 37-level grid.

4.2 CESM2-LENS2 (sigma, member `r10i1181p1f1`)

CESM2 PV computed on 11 sigma levels with surface pressure from the `PS` zarr store, bypassing all-NaN hybrid coefficients.

References

[1] Ertel, H. (1942). Ein neuer hydrodynamischer Wirbelsatz. *Meteorol. Z.*, 59, 277–281.

[2] Holton, J. R. & Hakim, G. J. (2013). *An Introduction to Dynamic Meteorology*, 5th ed. Academic Press.

[3] Skamarock, W. C. et al. (2012). A description of the Advanced Research WRF version 3. NCAR Technical Note NCAR/TN-475+STR.

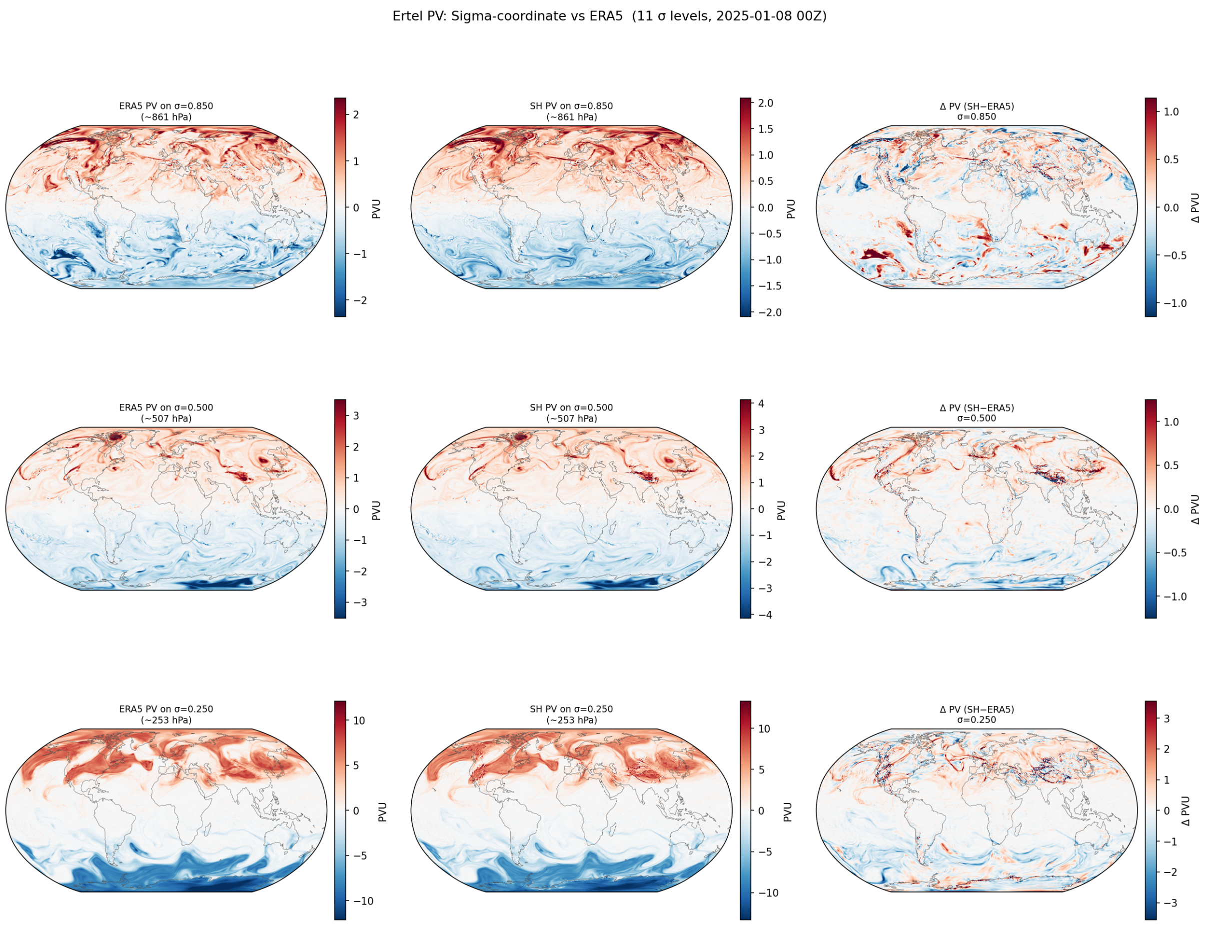

Figure 1: ERA5 PV validation on 11 sigma levels. Columns: ERA5 PV on σ, computed sigma PV, difference (SH − ERA5). Rows: $\sigma = 0.85$, 0.50, 0.25. (2025-01-08 00Z)

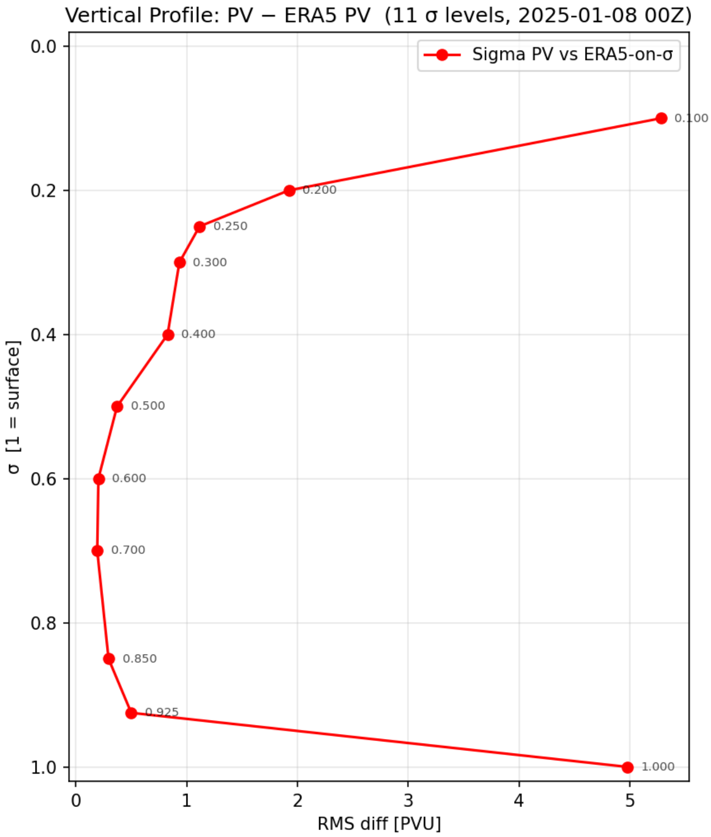

Figure 2: RMS difference profile: 11 sigma levels vs ERA5-on-σ. σ values annotated on each point.

../cesm2_compute/plots/cesm2le_pv_sigma.png

Figure 3: CESM2-LENS2 Ertel PV on 11 sigma levels (2010-01-01). $\sigma = 0.85$, 0.50, 0.25.